END USER COMPUTING.

BICT 1101

Lecture 6: Internet & World Wide Web

Internet

- It is the interconnection of computer networks all over the world so that users can share and exchange information and communicate with each other.
- It is worldwide collection of *networks* that connects millions of businesses, government agencies, educational institutions and individuals.

- Internet is a network made up of hundred of thousand of computers all over the world, with millions of users connected electronically by communication lines (telephone lines, fiber-optic cables and other links).
- *Internet is a huge network of computers all connected together.*

- A **computer network** is a connection of two or more computers for the purpose of communicating and sharing information and other resources.
- A **computer network** is a set of computers connected together for the purpose of sharing resources.
- Computers on a network can be connected through cables, telephone lines, radio waves, satellites, or infrared light beams or through wireless communication.

- The aim of connecting these computers is to enable sharing of resources such as printers, documents, computer programs, hard disk etc.
  - Common resource shared today is connection to the Internet.

Types of Computer Network

- Local Area Network (LAN)
  - An interconnection of computers and other devices in a limited geographical area.
- Wide Area Network (WAN)
  - Covers a large geographical area to provide national and international communication.

Purpose of LAN

- LANs allow:
  - Data transfer between computers
  - Access to data stored somewhere
  - Messages to be passed between computers
  - Shared use of peripherals e.g. printers
  - Shared use of computing resources

What is needed to connect to the Internet ?

- Hardware
  - I.e Computer, Modem, Dongle, Telephone line, Wireless infrastructure, cables, hub, switch, routers etc.
- Internet Service Providers (ISP)
  - **ISP** refers to a company that provides Internet services, for personal and business access to the Internet.
  - i.e MTL, Airtel, TNM, Globe Internet, Skyband etc..

What is needed to connect to the Internet ?

- Software
  - Web Client Software (Browsers) e.g. Internet Explorer, Mozilla Firefox, Google Chrome.

What can you do with the Internet ?

- Find information by using search engines
- File Transfer (download & upload files)
- Send and receive messages by electronic mail (email)
- School (eLearning)
- Electronic Trading (eCommerce)
  - Companies and their customers making use of networks as a means of advertising, buying and selling goods and services
- Teleworking or Telecommuting
  - Making use of networking technology to allow working at different location while keeping in touch with co-workers

What can you do with the Internet ?

- Tele and video conferencing
  - Groups of people conducting meetings via a telephone or video network.
- Publish information on your own web site
- Make airline reservations
- Get directions and draw maps
- International debates or discussion groups

- In order to communicate across the Internet, a computer must have a unique address.
- Every computer on the Internet has a unique numeric identifier, called an **Internet Protocol (IP) address**.
- An **IP address**, is a number that uniquely identifies each computer or device connected to the Internet.
- IP address usually consists of four groups of numbers, each separated by a period.
- The number in each group is between 0 and 255. For example, the numbers 72.14.207.99 and 192.168.10.33 are IP addresses.

- In addition to an IP address, most Internet hosts or servers have a Domain Name System (DNS) address which uses words.
  - A **domain name** is a unique name that identifies an internet resource such as a website
  - A **domain name** is the text version of an IP address.
- Numeric IP addresses are difficult to remember and use.
- Thus, Internet supports the use of a text name that represents one or more IP addresses.

- A domain name identifies the type of institution that owns the computer.
  - An Internet server owned by IBM might have the domain name ibm.com.
  - An Internet server owned by Mzuzu University have the domain name mzuni.ac.mw.

Internet Domains

Domain	Type of Organization	Example
.com	Business (commercial)	ibm.com (International Business Machines Corp.)
.edu	Educational	center.edu (Centre College, Danville, KY)
.gov	Government	Whitehouse.gov (The White House)
.mil	Military	Navy.mil (The United States Navy)
.net	Gateway or host (or business/commercial)	Mindspring.net (A regional Internet service provider)
.org	Other organization (typically nonprofit)	isoc.org (The Internet Society)

World Wide Web (www, web)

- **World Wide Web** is a system of connected documents on the Internet that can be searched for information about a particular subject.
- **World Wide Web** is a collection of electronic documents that are linked together like a spider web.
- **Web** is a collection of webpages found on the network of computers.
- **Web** is part or service of the Internet that uses text, images, sound, and video.
- These electronic documents are stored on computers called **servers** located around the world.

Web Page

- A web page is an electronic document or resource of information that can be accessed through a web browser.
- A web page is a page of information on the Internet about a particular subject that forms (a part) a website
- A web page is an electronic document on the world wide web.
- A web page is an electronic document written in a computer language called **HTML** (Hypertext Markup Language).

Web page

- Web pages are stored on web servers located around the globe.
- Web server is a computer system or program that finds and send web pages in response to requests from Internet users.
- A web page can contain text, graphics, video, animation, links to other pages, links to other files, and sound, as well as interactive features, such as data entry forms.

Hyperlink

- Hyperlink is a connection that allow users to move between two web pages on the Internet
- A web page contains links to other web pages as well.
- These links are called hyperlinks and
- Hyperlinks could be words, pictures or any graphics
- *Hyperlinks* connects users to other pages of related information.

Hyperlink

- Allows quick navigation through a website or to a new website.
- Mouse pointer will turn into a hand shape when moved over a link.
- With hyperlinks one can move through a vast amount of information by jumping from one web page to another.
- ***Navigation through pages of information is called surfing or browsing.***

Web Site

- A website is a set of pages of information on the Internet about a particular subject.
- A **website** is one or more web pages that relate to a common theme, such as a person, business, organization or a subject, such as sports.
- A website is a set of web pages consisting of text, audio and video.
- A web site is similar to a magazine with articles. It is a collection of web pages.

- A web page is an individual page that can be found on a web site.
- Most websites have a home page as their starting point, which functions as a table of contents for the site.

Web Browser

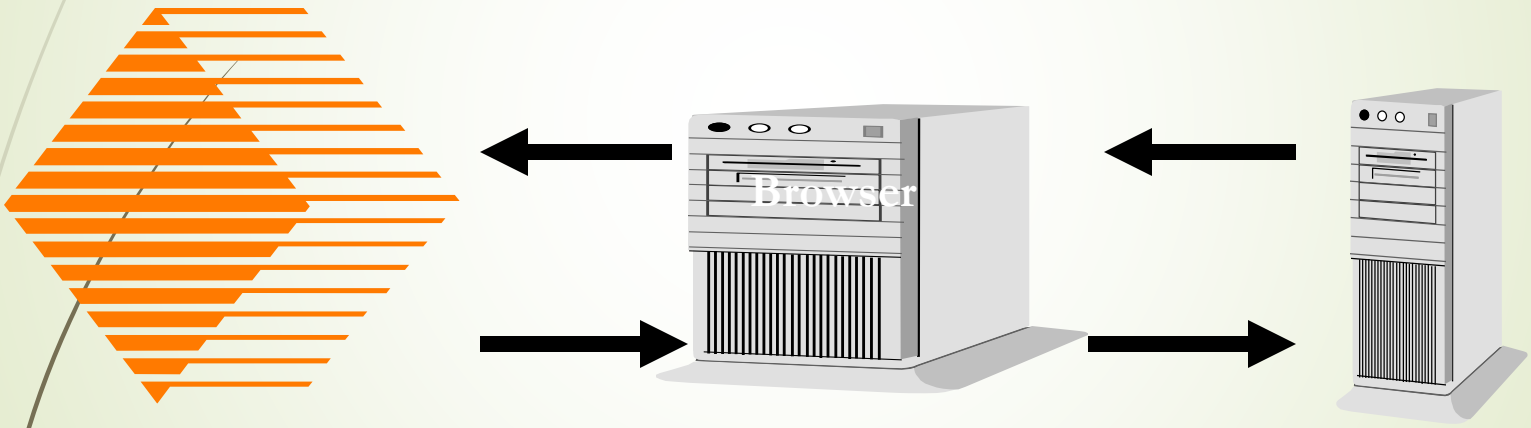
- **Web browser** is a software application used to locate, retrieve and display content on the World Wide Web, including Web pages, images and video.
- A web browser is a computer program that makes it possible for users to read information on the Internet.

- A web browser is the software that users need in order to find, retrieve, view, and send information over the Internet.
  - User enters web address (URL) into the web browser's address bar.
  - The browser communicates with the server to get the information requested by the user.
- Examples of Web Browsers are: Mozilla Firefox, Google Chrome, Apple Safari, Opera, Internet Explorer etc.

How the Web Works

5. User receives file displayed by the browser

4. Server sends requested files to browser to be interpreted



1. 1.User sends request

2. Browser interprets user's selection and makes request from appropriate server

3. Server accepts and processes request from browser

Web Server

- Web Server is a computer that provides world wide web services on the Internet.
- Web servers are computers that deliver (*serves up*) Web pages.
- A Web server is a system that delivers content or services to end users over the Internet.
  - A Web server consists of a physical server (hardware), server operating system (OS) and software used to facilitate HTTP communication.
  - A Web server is also known as an Internet server.

Uniform Resource Locator (URL)

- URL is the address of an Internet page on a Web site. E.g. http://www.mzuni.ac.mw
- URL is the address that defines the route to a file on the Web.
- URL indicates where a web page is located on the internet.
- URL is a location identifier associated with a particular webpage or file on the Internet.
- URL is typed into web browser's address bar to access web pages.

- URL contains protocol prefix, domain name, subdirectory name and file name. E.g.
  - http://www.mzuni.ac.mw/downloads/student_handbook.pdf
- URL specifies Internet address of a file stored on a host computer connected to the Internet.
- Every file on the Internet, has a unique URL.
- Web browsers use URL to retrieve a file from a web server in which it resides.
- This file is then displayed on the monitor connected to the user's local machine.

URL format

- Each part of the URL directs the browser to the correct web page.

http://library.sullivan.edu/datainfo.html

Hypertext Transfer Protocol (HTTP)

Network Domain (domain name)

Filename (path)

Protocols

- A set of rules that enable two devices to connect and transmit data to one another.
- A computer language allowing computers that are connected to each other to communicate
- Protocols determine how data is transmitted between computing devices and over networks.

TCP /IP

- i)Transmission Control Protocol.
  - This is the standard protocol computers utilize to communicate with one another over a network.
- ii) Internet Protocol
  - This a protocol that enable computers to identify each other over the Internet

Protocols

- **HTTP** – Hypertext Transfer Protocol
  - HTTP is the set of rules for transferring files (text, graphic images, sound, video, and other multimedia files) on the World Wide Web.
  - A protocol for informing web browsers how to display information found on a particular webpage.
  - The main protocol used by the World Wide Web to define how messages are formatted and transmitted.
  - All Web Pages on World Wide Web use HTTP, making them accessible to all other Web computers.

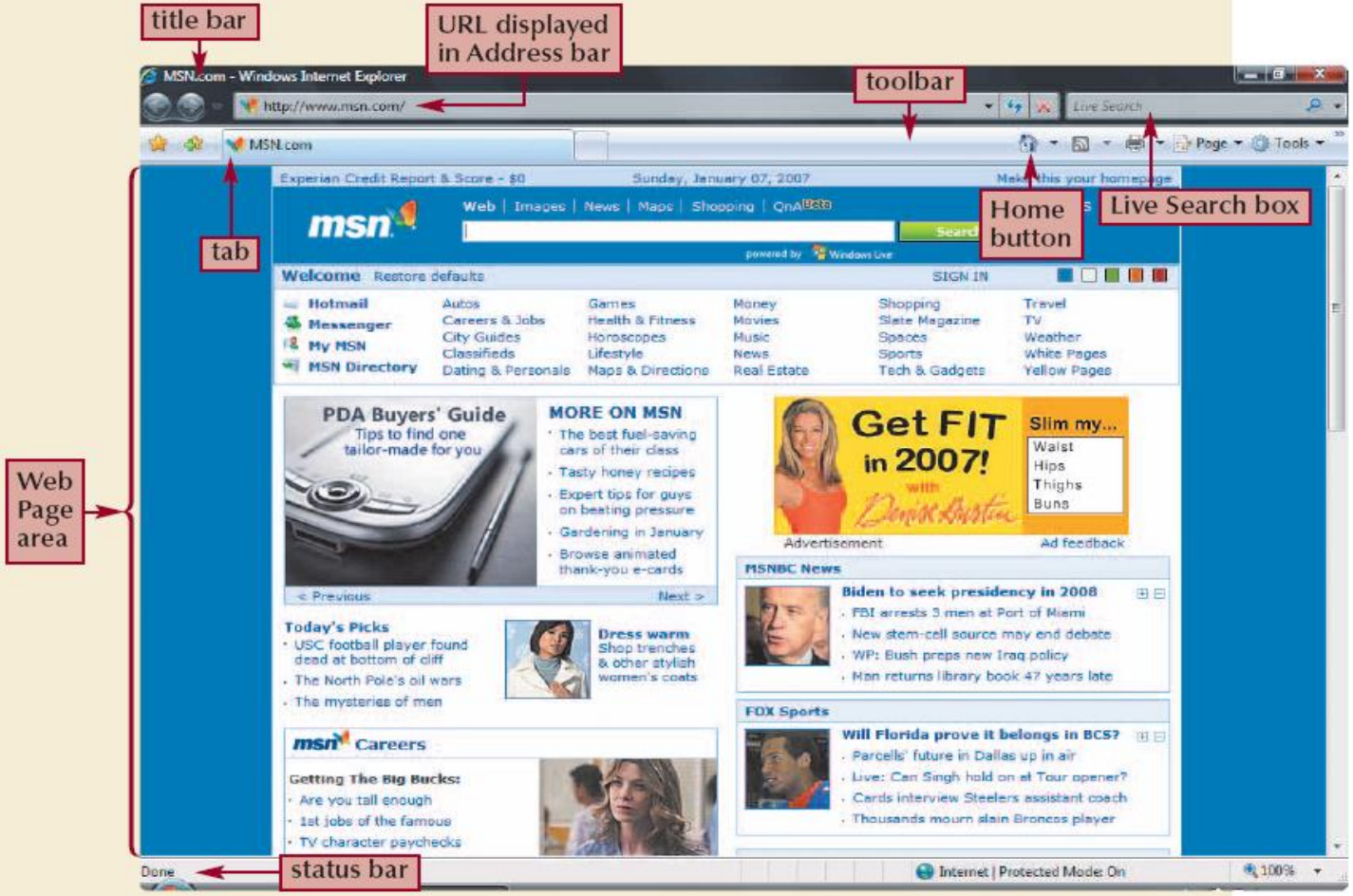
- **FTP** – File Transfer Protocol.
  - This is a protocol computers use to send files to one another.
  - You may either upload or download files using FTP
    - **Upload** – To forward a file to a server for later access over the Internet.
    - **Download** – To copy a file from an Internet server to recordable media, such as a hard drive, CD, DVD or flash disk.
- **Protocols** involved in sending and receiving emails:
  - Post Office Protocol 3 (POP3)
  - Internet Message Access Protocol (IMAP)
  - Simple Mail Transfer Protocol (SMTP)

Web Browsers: Microsoft Internet Explorer

- Microsoft Internet Explorer is the one example of web browser.
- It comes with windows operating system.
- When you start Internet Explorer, Internet Explorer's main browser window opens automatically
- To display a web page you want, you need to know the website address of the web page you want to view

Microsoft Page

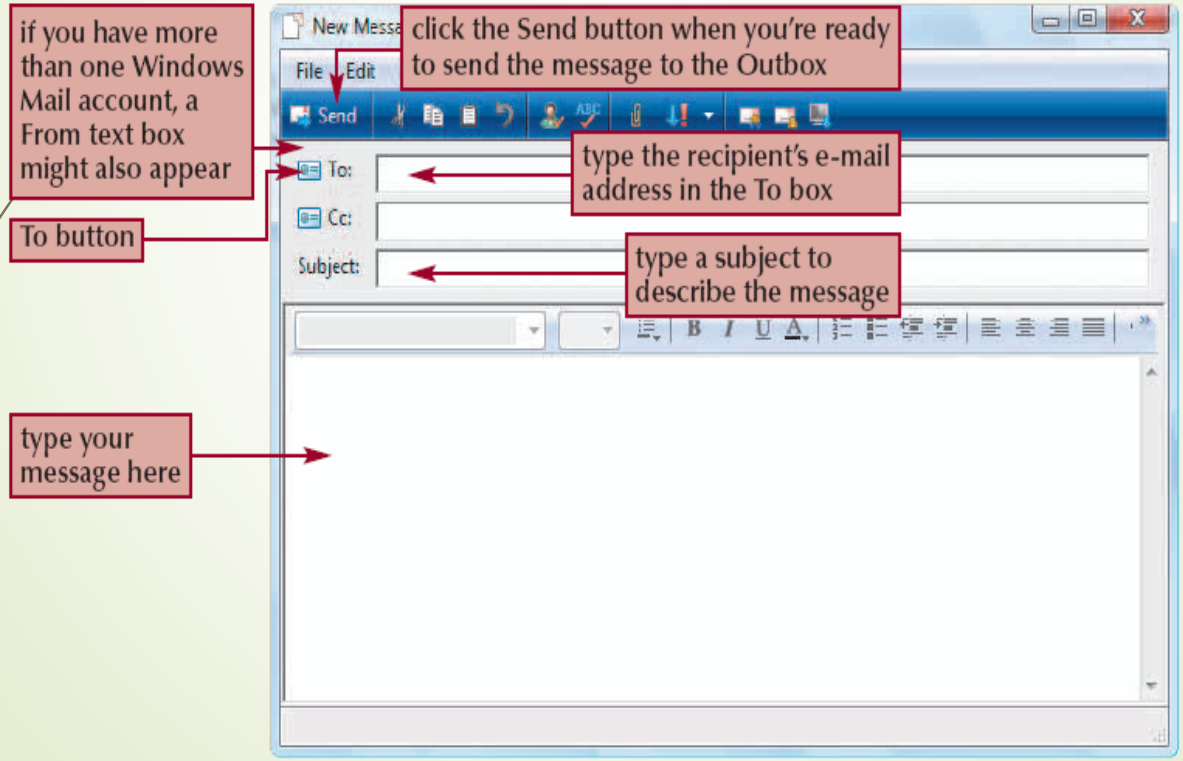
Internet Explorer window — Figure 4-3



Getting Started with Microsoft Internet Explorer

- Click the **Start button** on the taskbar, and then click **Internet Explorer**

New Message window — Figure 4-19



How to Access Web Pages

- Three ways to access Web pages:
  - Type the Uniform Resource Locator (URL) in the address box.
  - Click on hyperlinks on a web page.
  - Use a search tool such as search engine or web directory.

How to Access Web Pages: Type the URL in the Address Box

- Type the Uniform Resource Locator (web address) in the browser address box.
- Click GO (or press ENTER key on the keyboard).
- The URL must be correct (no misspellings, extra spaces, etc.) to go to the correct web page.

Opening a Page on the Web

- Use Back and Forward buttons to navigate through the pages you have visited
- To open a Web page you visited in the last few days, use History list or Bookmarks.
- To open a new Web page without closing the one you're currently viewing, use tabbed browsing
  - If you have two or more tabs open, use Quick Tabs button to display a thumbnail of each Web page

Changing Home Page

- Home Page is the first page which appears every time you start Internet Explorer.
- You can display or even change your home page.
- To display the web page you want and set as your home page
  - Click on Tools in the menu bar. The Tools Menu will appear.
  - Click on Internet Options
  - Internet Options dialog box will appear
  - The page displays the address of your current home page

- To set the web page displayed on your screen as your new home page, click on use current button
- The address box displays the address of your new home page
- Click on the OK button

Email (Electronic mail)

- Email is a method of exchanging digital messages from an author to one or more recipients.
- Email is like sending a letter, only you send the message over the internet and it gets to the other person's email account within seconds.
- If you have an email address, you can email anyone else who has an email address.
- Once you've got an email account, you can store your emails on the internet, or store them on a computer, laptop or another internet-connected device.

- **Key benefits and features of using email**
- It's quick – your recipient receives your email as soon as they go online and collect their mail.
- It's secure.
- It's low cost.
- Photos, documents and other files can be attached to an email, so that more information can be shared.
- One email can be sent to more than one recipient at a time.

- If your email is stored on the web, it's called 'webmail'.
- To access a webmail account, you get onto the internet, go to the website where your webmail is hosted and 'log in' or 'sign in' with your username and password.

Email addresses

- Email addresses use a standard formula.
- They are always separated into two parts by the @ symbol.

- The first part of the email address identifies the user.
- You can use real names for the first part of your email address, or you could use a nickname that your friends and family will recognise.
- The second part of an email address comes from your webmail provider or the internet service provider you use.
- This gives a standard formula for all email addresses which will look something like firstname.surname@somewhere.com.au or yournickname@somewhere.com.au

Using email

- All email accounts have the same basic features.
- They will have an '**Inbox**' for the emails you've received.
- There will also be a '**Sent**' items folder where the emails that you've sent to other people can be saved.
- And there will be an '**Address book**' or '**Contacts**' list where you can save other people's email addresses.

Sending emails

- You can create new emails to send to people.
- Creating a new email might be called 'compose', 'create', or 'new email'.
- To send an email, you type the email address or email addresses into the 'to' field.
- There is also a 'cc' field and a 'bcc' field showing that you can use to send copies of this email to other people.
- In the subject field, you tell the other person what the email is about and then write the email in the body field.
- You can also attach files to emails including photos and documents.

- When you click **Send**, the email is sent to the other person's email account over the internet.
- They'll get your email next time they log into their account.
- If the email didn't reach the other person, you'll get an automated email telling you that the email was undeliverable.
- This could be because the email address was spelled incorrectly, you accidently included spaces, or sometimes it's because the other person's inbox is full and can't receive any more emails until they delete some emails to make space for new ones.

Replying to and forwarding emails

- If someone emails you, you can reply to them by selecting the '**Reply**' button.
- You then write back to them and then select '**Send**'.
- And if you receive an email that you'd like to share with someone else, you can forward the email to other people by selecting the '**Forward**' button, putting in other people's email addresses, typing in a message and selecting 'send'.

Tips on emails

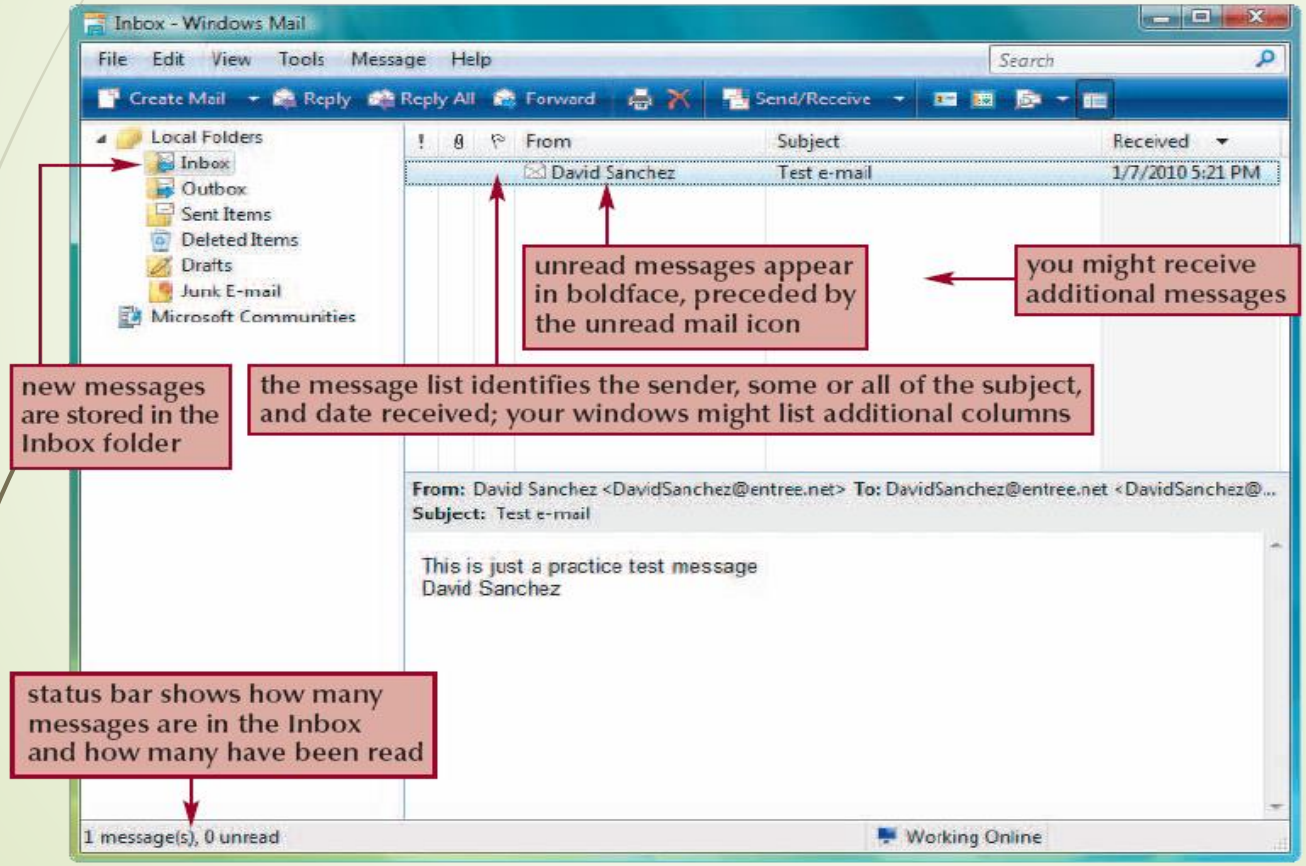
- Respond only as necessary
- State your subject clearly
- Don’t e-mail sensitive or confidential information
- Be concise and direct
- Avoid abbreviations
- Don’t use all capital letters
- Pause and reread before sending

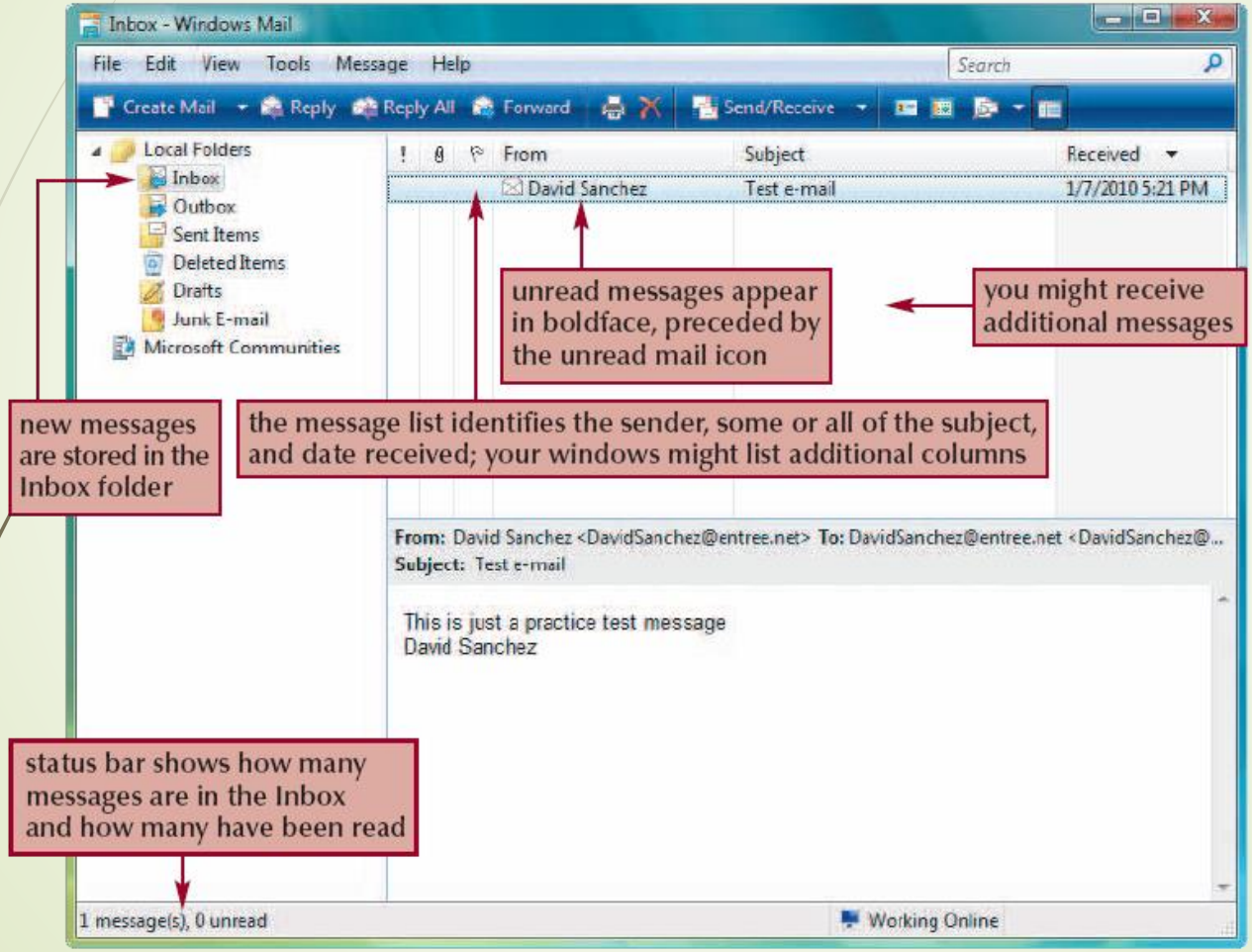
Creating and Sending Email Message

- Click Create Mail button on the toolbar
- Enter e-mail address of the recipient in the To box
- Click Subject box and then type the subject of the message
- Click the message area and then type the content of the message
- Click Send button.

Creating and Sending an E-mail Message

Inbox folder **Figure 4-21**



Receiving an E-mail Message

Inbox folder **Figure 4-21**